Evo2 restricted-scan benchmark

Historical August 2026 run. The frozen Evo2-7B probe produced fewer benchmark-unmatched peaks than the compared CNN at all listed recall targets. This reversed the earlier AUPRC-only decision to drop Evo2; it now belongs to the deployed trio.

Method

The weighted probe selects logreg/RF by validation AP, refits train+validation (47,137 rows) and evaluates test. Historical logreg validation/test AP is 0.7007/0.7026. Saved-feature recovery later reproduced weighted test AP 0.70258010/AUROC 0.89534660 on 3,053 rows/547 positives; it did not rerun extraction or the scans below.

The current frozen recipe uses 81-bp windows, mean-pooled blocks.28.mlp.l3 and 4096 float32 features. Exact historical source revision/layer remains unresolved. Model documentation identifies the recovered data.

Restricted regions span 300 bp upstream/50 bp into annotated genes, by strand. Step-10 windows overlap these regions. Same-contig/strand windows expanded ±40 bp cover catalogue TSS; unmatched starts merge at gaps ≤81 bp. Precision proxy is covered TSS/(covered TSS+unmatched peaks), not matched-call precision. Historical "FP" means unmatched, not experimentally false.

The CNN step-1 track was subsampled to step 10. Equal stride does not prove equal grids; the standalone Evo2 jobs omit `--align-grid`, which deployment adds. Both benchmark strains supplied silver/negative training windows while published TSS labels were withheld.

Curated classification

Model	Test AUPRC
ProkBERT (fine-tuned)	0.884
NT-LoRA (fine-tuned)	0.854
CNN-v2 (from scratch)	0.848
CNN-v1 (from scratch)	0.804
Evo2 frozen + logreg	**0.703**

Task-trained rows are earlier T21_final/plain-0.6 checkpoints; Evo2 uses T15_xs, where deployed CNN-v2 AP is 0.726. These numbers do not establish a controlled ranking. Test prevalence is about 18%, not 50%.

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GCF_000143685.1/NC_014328.1, 1,245 published TSS; historical restricted-window coverage baseline 0.70%, not promoter prevalence.

Recall	Evo2 (step 10)	cnn_v2-xs (step 10)	Δ precision	cnn_v2-xs (step 1, ref)
0.10	**71.0%** (51 FP)	63.1% (73 FP)	+7.9	71.6%
0.50	**58.0%** (451 FP)	48.2% (670 FP)	+9.8	51.9%
0.70	**44.8%** (1,074 FP)	38.3% (1,403 FP)	+6.5	—
0.90	**27.1%** (3,012 FP)	24.8% (3,391 FP)	+2.3	—

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GCF_000484505.1/NC_022592.1, 2,885 published TSS; historical coverage baseline 1.70%.

Recall	Evo2 (step 10)	cnn_v2-xs (step 10)	Δ precision
0.10	**85.8%** (48 FP)	81.2% (67 FP)	+4.6
0.50	**71.4%** (581 FP)	67.1% (708 FP)	+4.3
0.70	**59.7%** (1,361 FP)	57.3% (1,507 FP)	+2.4
0.90	**46.9%** (2,939 FP)	40.2% (3,864 FP)	+6.7

Interpretation

Compare members within a genome. Catalogue density/completeness differs; absolute precision is not transferable target accuracy. The step-1 CNN reference and later paired-grid numbers are separate runs: paired Evo2 gives 69.4/58.4% and 87.6/72.5%, while the trio gives 74.4/61.6% and 87.0/76.6%.

These results motivate Evo2's inclusion, not a causal pretraining explanation or promoter confirmation. Original run checks are not preserved as a new audit; source inspection alone does not replace execution. The probe retains the 888 wrong-assembly training positives. Historical runtime around one GPU-hour depends on hardware/batching/coverage. Removing `--regions` from a benchmark cannot reconstruct unscored windows.

Pipeline gives reproduction prerequisites; interpretation separates ranks/motifs from biological claims.